8.4 Generate Statement

- The generate statements are concurrent statements with embedded internal concurrent statement, which can be interpreted as a circuit part.
- Two types of generated statements:
  - for generate statement: used to create a circuit by replicating the hardware part
  - conditional or if generate statement: used to specify whether or not to create an optional hardware part.

8.4.1 For Generate Statement

- Many digital circuits can be implemented as a repetitive composition of basic building blocks, exhibiting a regular structure, such as a one-dimensional cascading chain, a tree-shaped connection or a two-dimensional mesh.
- For generate statement syntax

```
gen_label:      -- mandatory to identify to this
   (必须)       -- particular generate statement
for loop_index in loop_range generate
    concurrent statements;
    -- describe a stage of the  iterative circuit
end generate;
```

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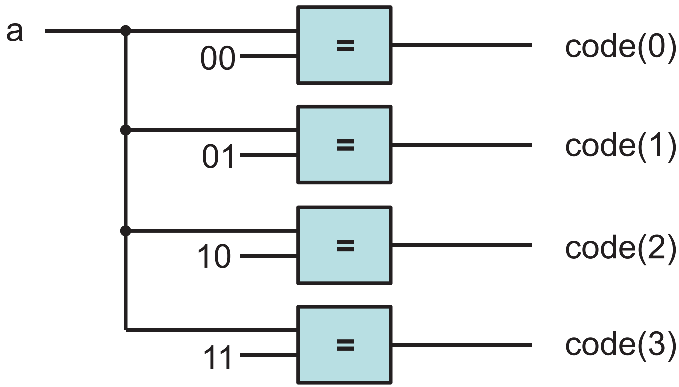
- The loop_range has to be static. It is normally specified by the width parameters.

Example: Binary decoder

- A binary n-to-2^n decoder is circuit that asserts one of the 2^n possible output signal according to an n bit input signal.
- One way to view the binary decoder is to treat each bit of the decoded output as the result of a constant comparator.



分别跟00，01，10，11进行比较

code(i) <= ‘1’ **when** a = std_logic_vector(unsigned(i)) **else** ‘0’;

Parameterized binary decoder using a for generate statement

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity bin_decoder is
    generic(WIDTH: natural);
    port(
        a: in std_logic_vector(WIDTH–1 downto 0);
        code: out std_logic_vector(2**WIDTH–1 downto 0)
    );                                          2^WIDTH - 1
end bin_decoder;

architecture gen_arch of bin_decoder is
begin
    comp_gen:   必须有label
    for i in 0 to (2**WIDTH–1) generate
        code(i) <= ‘1’ when i = to_integer(unsigned(a)) else
                     ‘0’;          generate语句中的电路是同时生成的
    end generate;
end architecture gen_arch;
```

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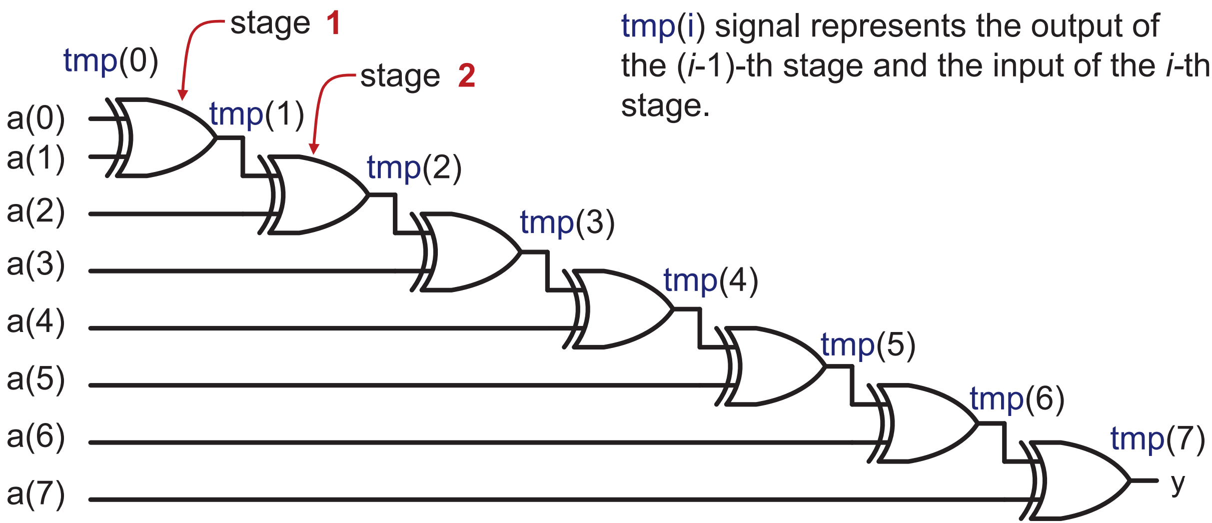
Example: Reduced-xor circuit



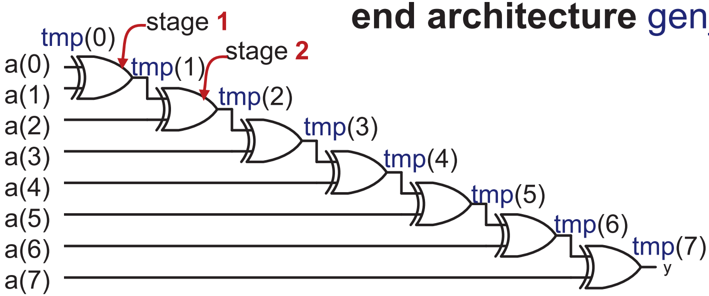
tmp(i) signal represents the output of the (*i*-1)-th stage and the input of the *i*-th stage.

tmp(i+1) <= tmp(i) **xor** a(i+1)

tmp(i) <= tmp(i-1) **xor** a(i)

Parameterized reduced-xor circuit using a for generate statement

```
architecture gen_linear_arch of reduced_xor is
    signal tmp: std_logic_vector(WIDTH–1 downto 0);
begin
    tmp(0) <= a(0);
    xor_gen:
        for i in 1 to (WIDTH–1) generate
            tmp(i) <= a(i) xor tmp(i–1);
        end generate;
    y <= tmp(WIDTH-1);
end architecture gen_linear_arch;
```



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- In an iterative structure, the boundary stages interface to the external input and output signals, and sometimes their connections are different from the regular blocks.

8.4.2 Conditional Generate Statement

- The conditional generate statement is used to specify an optional circuit that can be included or excluded in the final implementation.
- Conditional generate statement syntax

```
gen_label:      -- mandatory
if boolean_exp generate -- boolean_exp must be static
     concurrent statements;
end generate;
```

- There is no else branch in conditional generate statement.
- If we want to include one of the two possible circuits in an implementation, we must use two separate if generate statements.

Reduced-xor circuit revisited

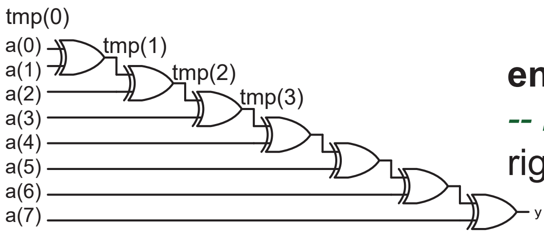
- One common use of the conditional generate statement is to describe the “irregular” stages in a for generate statement.
- For example, two statements

```
tmp(0) <= a(0);
y <= tmp(WIDTH-1);
```

  are used to rename the input and output signals in the for generate statement examples.
- To eliminate these statements, we can use conditional generate statements inside the for generate statement.

Parameterized reduced-xor circuit with a conditional generate statement

```
tmp(0) <= a(0);
xor_gen:
    for i in 1 to (WIDTH–1) generate
        tmp(i) <= a(i) xor tmp(i–1);
    end generate;
 y <= tmp(WIDTH-1);
```



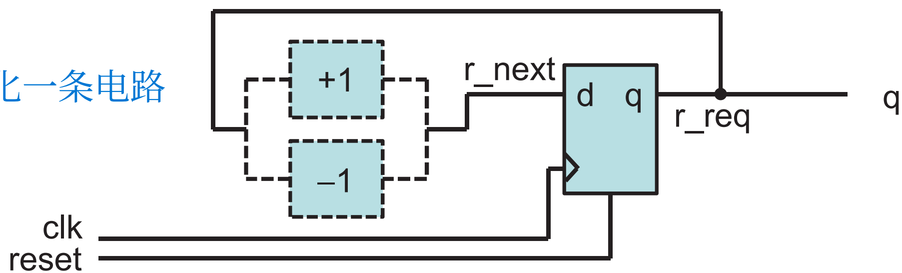

```
architecture gen_if_arch of reduced_xor is
    signal tmp: std_logic_vector(WIDTH–2 downto 1);
begin
    xor_gen:
    for i in 1 to (WIDTH–1) generate
        -- leftmost stage
        left_gen: if i = 1 generate
            tmp(i) <= a(i) xor a(0);
        end generate;
        -- middle stage
        middle_gen: if (i>1) and (i<(WIDTH –1)) generate
            tmp(i) <= a(i) xor tmp(i–1);
        end generate;
        -- rightmost stage
        right_gen: if i = (WIDTH –1) generate
            y <= a(i) xor tmp(i–1);
        end generate;
    end generate;
end architecture gen_if_arch;
```

for generate + if generate

Example: Up-or-down free-running binary counter

- An up-or-down binary counter is a counter that can be instantiated in a specific mode.
- Note that the "**or**" here means that only one mode of operation, either counting up or counting down but not both, can be implemented in the final circuit.

只实例化一条电路



- We use the UP generic as the feature parameter to specify the desired mode.

Up-or-down free-running binary counter

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity up_or_down_counter is
    generic(WIDTH: natural;  UP: natural);
    port(clk, reset: in std_logic;
        q: out std_logic_vector(WIDTH–1 downto 0)
    );
end up_or_down_counter;

architecture arch of up_or_down_counter is
    signal r_reg, r_next: unsigned(WIDTH–1 downto 0);
begin
    -- register
    process (clk, reset)
    begin
        if (reset = ‘1’) then
            r_reg <= (others => ‘0’)
        elsif (clk’event and clk=‘1’) then
            r_reg <= r_next;
        end if;
    end process;
```

消耗更少的硬件资源

```
    -- next-state logic
    inc_gen: -- incrementor
    if UP = 1 generate
        r_next <= r_reg + 1;
    end generate;
    dec_gen: -- decrementor
    if UP /= 1 generate
        r_next <= r_reg – 1;
    end generate;
    q <= std_logic_vector(r_reg);  -- output logic
end architecture arch;
```

Up-and-down free-running binary counter

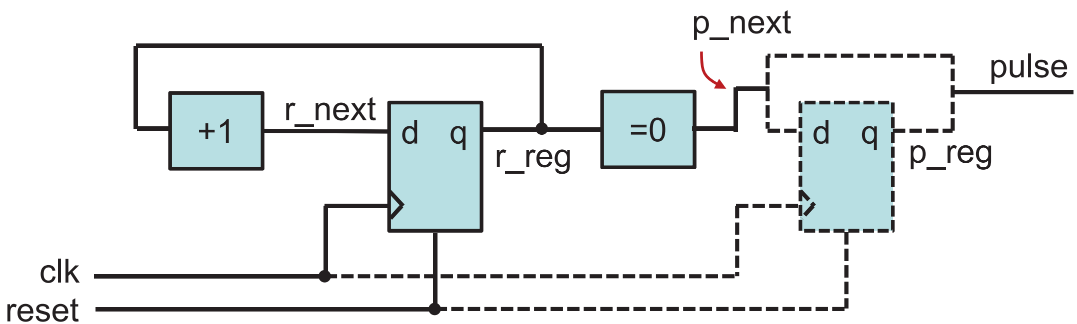
```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity up_and_down_counter is
    generic map (WIDTH: natural)
    port map(clk, reset: in std_logic; mode: in std_logic;
        code: out std_logic_vector(2**WIDTH–1 downto 0)
    );
end up_and_down_counterr;
```

```
architecture arch of up_and_down_counter is
    signal r_reg, r_next: unsigned(WIDTH–1 downto 0);
begin
    -- register
    process (clk, reset)
    begin
        if (reset = ‘1’) then
            r_reg <= (others => ‘0’)
        elsif (clk’event and clk=‘1’) then
            r_reg <= r_next;
        end if;
    end process;
    -- next-state logic
    r_next <= r_reg + 1 when mode =‘0’ else
              r_reg -1;
    -- output logic
    q <= std_logic_vector(r_reg);
end architecture arch;
```

Counter with an optional output buffer

- An output buffer can remove glitches from the signal.
- Since the buffer is only needed for certain application, it will be convenient to include the buffer as an optional part of the circuit.



Counter with an optional output buffer

不能单独存在，**要在**
其他地方实例化，**不然**
WIDTH**和**BUFF的值都
不确定没法生成电路

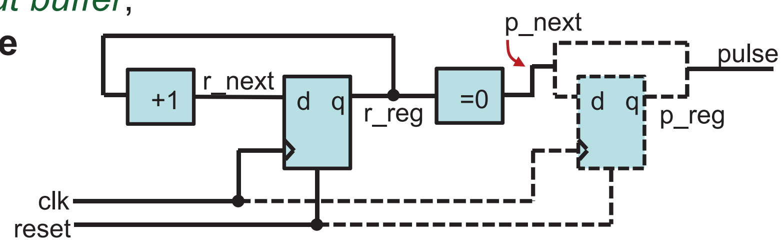
```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity op_buf_counter is
    generic(WIDTH: natural; BUFF: natural);
    port(clk, reset: in std_logic;
        pulse: out std_logic);
end op_buf_counterr;

architecture arch of op_buf_counter is
    signal r_reg, r_next: unsigned(WIDTH–1 downto 0);
    signal p_reg, p_next: std_logic;
begin
    -- register
    process (clk, reset)
    begin
        if (reset = ‘1’) then r_reg <= (others => ‘0’)
        elsif (clk’event and clk=‘1’) then r_reg <= r_next;
        end if;
    end process;
```

```
-- next-state logic
r_next <= r_reg + 1;
-- output logic
p_next <= '1' when r_reg = 0 else '0';
buf_gen:
if BUFF = 1 generate
    process (clk, reset)
    begin
        if (reset = '1') then p_reg <= '0'
        elsif (clk'event and clk = '1') then p_reg <= p_next;
        end if;
    end process;
    pulse <= p_reg;
end generate;
no_buf_gen: -- without buffer;
if BUFF /= 1 generate
    pulse <= p_next;
end generate;
end architecture arch;
```

整个process是
一个并发语句



8.5 Comparison

- To create a circuit with a selectable feature:
  - use conditional generate statement
  - a full-featured circuit with some input control signal connected to constant values to permanently enable the desired feature
  - use the configuration construct 调用entity**的不同**architecture
- Assume we need a 16-bit up counter in a design.

```
count16up: up_or_down_counter
    generic map (WIDTH => 16, UP =>1)
    port map (clk => clk, reset => reset, q=>q);
```

or

```
count16up: up_and_down_counter
    generic map (WIDTH => 16)
    port map (clk => clk, reset => reset, mode => '1', q=>q);
```

Difference

- The up-or-down counter instance
  - creates a circuit with only the needed features.
  - The selected portion of code is passed to the synthesis stage, i.e., the synthesis software only needs to synthesize the selected portion.
- The up-and-down counter instance
  - creates a circuit that consists of all features and uses an external control signal to selectively enable a portion of the circuit.
  - The entire VHDL code is passed to synthesis stage. The synthesis software eliminates the unused portion through logic optimization.
- In general, use of the feature parameters and conditional generate statement is better than the full-featured approach.

- The selected hardware creation can also be achieved by configuration where multiple architecture bodies are constructed, each containing a specific feature, e.g., architectures up_arch and down_arch of the same entity updown_counter, for counting up and counting down, respectively.
- And the following instantiation can be used to select the counting up circuit.

```
count16up: work.updown_counter(up_arch)
    generic map(WIDTH =>16)
    port map (clk => clk, reset => reset, q => q);
```

Up-or-down counter with two architecture bodies

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity updown_counter is
      generic(WIDTH: natural);
      port(clk, reset: in std_logic;
            q: out std_logic_vector(WIDTH–1 downto 0)
      );
end updown_counter;

architecture up_arch of updown_counter is
      signal r_reg, r_next: unsigned(WIDTH–1 downto 0);
begin
      -- register
      process (clk, reset)
      begin
            if (reset = '1') then r_reg <= (others => '0')
            elsif (clk'event and clk='1') then r_reg <= r_next;
            end if;
      end process;
```

```
    -- next-state logic
    r_next <= r_reg + 1;
    -- output logic
    q <= std_logic_vector(r_reg);
end architecture up_arch;

architecture down_arch of updown_counter is
    signal r_reg, r_next: unsigned(WIDTH–1 downto 0);
begin
    -- register
    process (clk, reset)
    begin
        if (reset = ‘1’) then r_reg <= (others => ‘0’)
        elsif (clk’event and clk=‘1’) then r_reg <= r_next;
        end if;
    end process;
    -- next-state logic
    r_next <= r_reg – 1;
    -- output logic
    q <= std_logic_vector(r_reg);
end architecture down_arch;
```

- Conversely, we can merge the logic from several architecture bodies into a single body and use a feature generic and conditional generate conditions to select the desired portion.
- There is no rule about when to use a feature parameter and when to use a configuration construct. In general,
  - code with a feature parameter is more difficult to develop and comprehend, but on the other hand, if we use a separate architecture body for each distinctive feature, the number of architecture bodies will grow exponentially and becomes difficult to manage.
  - when a feature parameter leads to significant modification or addition of the no-feature codes and starts to make the code incomprehensible, it is probably a good idea to use separate architecture bodies and the configuration construct.

8.6 For Loop Statement

- The for loop statement is a sequential statement and is the only sequential loop construct that can be synthesized.

must be static

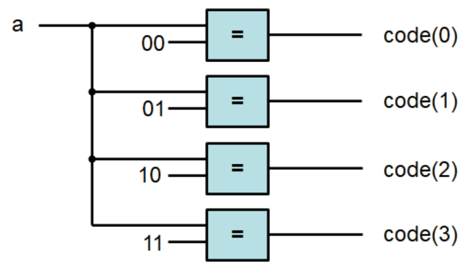
for index **in** loop_range **loop**
sequential statements;
end loop;

- The basic way to synthesize a for loop statement is to unroll or flatten the loop. Unrolling a loop means to replace the loop structure by explicitly listing all iterations.

Example: Binary decoder

- The code is very similar to the for generate version **Slide 273**

```
architecture loop_arch of bin_decoder is
begin
   process (a)
   begin
      for i in 0 to (2**WIDTH–1) loop
         if i = to_integer(unsigned(a)) then code(i) <= '1';
         else code(i) <= '0';
         end if;
      end loop;
   end process;
end architecture gen_arch;
```



Example: Reduced-xor circuit

- For loop version: **Slide 259**
- For generate version: **Slide 275**

Example: Priority Encoder

- Recall that a signal can be assigned with multiple times inside process and only the last assignment takes effect.
- A priority encoder is a circuit that returns the binary code of the highest-priority request.
- Assume that the input is an array of r(WIDTH-1 **downto** 0), and r(WIDTH-1) has the highest priority.

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numberic_std.all;
use work.util_pkg.all;

entity prio_encoder is
    generic(WIDTH: natural);
    port(r: in std_logic_vector(WIDTH-1 downto 0);
        bcode: out std_logic_vector(log2c(WIDTH)-1 downto 0);
        valid: out std_logic);          计算以2为底的对数
end prio_encoder;
```

```
architecture loop_linear_arch of prio_encoder is
    constant B: natural := log2c(WIDTH);
    signal tmp: std_logic_vector(WIDTH-1 downto 0);
begin
    process (r) --binary code
    begin
        bcode <= (others => '0');
        for i in 0 to (WIDTH-1) loop    从低位向高位，高位的结果会把低位的结果覆盖
            if r(i) = '1' then
                bcode <= std_logic_vector(to_unsigned(i, B));
            end if;
        end loop;
    end process;

    process(r, tmp) -- reduced – or circuit
    begin
        tmp(0) <= r(0);
        for i in 1 to (WIDTH – 1) loop
            tmp(i) <= r(i) or tmp(i-1);
        end loop;
    end process;
    valid <= tmp (WIDTH-1);
end architecture loop_linear_arch;
```

8.7 Comparison

- Both the for generate and for loop statements are used to describe replicated structures.
- For generate statement:
  – can only use concurrent statements.
  – start a design with a conceptual diagram of a few stages; the diagram is used to identify the basic building block and connection pattern, and then the code of the loop body is derived.
- For loop statement:
  – can only use sequential statements.
  – the body of the loop statement can be more general and versatile.
  – may lead to unnecessarily complex implementation or even an unsynthesizable description.